

Mathematical Foundations of Reinforcement Learning and Online Decision Making

Lecture 4

Markov Decision Processes (MDPs) and Dynamic Programming (Part 2)

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Overview

- 1 Markov Reward Process - Average Reward
- 2 Value Iteration for Average Reward MRP
- 3 Markov Decision Process
- 4 Linear Programming Method

Markov Reward Process

A Markov chain is a pair $(\mathcal{S}, P, R)$ of

- 1 **States:** $\mathcal{S}$
- 2 **Transitions:** $P : \mathcal{S} \times \mathcal{S} \rightarrow [0, 1]$ where

$$\sum_{s' \in \mathcal{S}} P(s, s') = 1 \quad \text{for all } s \in \mathcal{S}.$$

- 3 **Reward function:** $R : \mathcal{S} \rightarrow \mathbb{R}$

Markov Reward Process - Trajectory

A trajectory (realization) of $(\mathcal{S}, P, R)$ is a sequence

$$S_0, R_0, S_1, R_1, S_2, R_2 \dots, S_t, R_t \dots$$

where

$$\begin{aligned} P(S_{t+1}=s' | S_t=s, S_{t-1}=s_{t-1}, \dots, S_0=s_0) &= P(S_{t+1}=s' | S_t=s) \\ &= P(s, s') \end{aligned}$$

and

$$R_t = R(S_t) \quad [\text{or } R_t \sim R(S_t)]$$

Markov Reward Process - Value of a State

Criterion 1: Discounted Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(S_t) \mid S_0 = s \right]$$

Criterion 2: Stochastic Shortest Path Problem

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} R(X_t) \mid S_0 = s \right]$$

Criterion 3: Average Cost per Stage Problem

$$V(s) = \mathbb{E} \left[\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T R(S_t) \mid S_1 = s \right]$$

Markov Reward Process - Average Reward

Average Reward per Stage Problem

$$V(s) = \mathbb{E} \left[\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T R(S_t) \mid S_1 = s \right]$$

Difficulties:

- Less intuitive one-step Bellmann function
- Not necessary contractive
- All time steps as important (slow convergence)
- Others

Markov Reward Process - Average Reward

Average Reward per Stage Problem

$$V(s) = \mathbb{E} \left[\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T R(S_t) \mid S_1 = s \right]$$

Assumption

The Markov chain has a single recurrence class.

- Does not need to be irreducible or aperiodic.
- Can derive similar results for general Markov chains, but more complicated.

Markov Reward Process - Average Reward

Gain

The average reward is same for all states, for all $s \in \mathcal{S}$

$$g = \mathbb{E} \left[\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T R(S_t) \mid S_1 = s \right]$$

We proved this in HW1 for Markov chains that were irreducible or aperiodic.

Assumption

The Markov chain has a single recurrence class.

Does not need to be irreducible or aperiodic.

Markov Reward Process - Average Reward

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We proved this in HW1 for Markov chains that were irreducible or aperiodic.

Markov Reward Process - Average Reward

Theorem

$$g = \sum_{s \in \mathcal{S}} r(s) \pi(s)$$

where $\pi : \mathcal{S} \rightarrow \Delta_{\mathcal{S}}$ is a stationary distribution of P , i.e. satisfying $\pi^T = \pi^T P$.

Markov Reward Process - Average Reward

Bias

$$h(s) = \mathbf{E} \left[\sum_{k=1}^{\infty} r(S_k) - g \mid S_1 = s \right].$$

If the limit does not exist then we can use the Cesaro limit

$$\text{C-lim}_{k \rightarrow \infty} a_k = \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=1}^K a_k$$

instead.

Markov Reward Process - Average Reward

Theorem

We have that

$$ge + (I - P)h = r$$

where e is a vector of all ones, and I is the identity matrix.

Can we solve the linear system?

Markov Reward Process - Average Reward

Theorem

We have that

$$ge + (I - P)h = r$$

where e is a vector of all ones, and I is the identity matrix.

Can we solve the linear system?

No, generally has many solutions, but we know what g , we fix one component of h , e.g. $h(s_1) = 0$

Gain and bias: a two-state example

Markov reward process

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}, \quad r = \begin{pmatrix} 2 \\ 0 \end{pmatrix}.$$

The stationary distribution is

$$\mu = \left(\frac{2}{3}, \frac{1}{3} \right),$$

so the gain is

$$g = \mu^\top r = \frac{4}{3}.$$

Bias equation

$$g\mathbf{1} + h = r + Ph.$$

Fixing $h(2) = 0$ gives

$$h(1) = \frac{10}{3}, \quad h(2) = 0.$$

Thus, for large T ,

$$\mathbb{E}_1 \left[\sum_{t=0}^{T-1} r_t \right] \approx \frac{4}{3}T + \frac{10}{3},$$

$$\mathbb{E}_2 \left[\sum_{t=0}^{T-1} r_t \right] \approx \frac{4}{3}T.$$

Markov Reward Process - Average Reward

Theorem - Relation with Discounted MDPs

For discount $\lambda \in [0, 1)$, let $v_\lambda : \mathcal{S} \rightarrow \mathbb{R}$ be the discounted value

$$v_\lambda(s) = \mathbb{E} \left[\sum_{k=0}^{\infty} \lambda^k R(S_k) \mid S_0 = s \right].$$

Then

$$v_\lambda = (1 - \lambda)^{-1} g + h + f(\lambda)$$

where $\lim_{\lambda \rightarrow 1^-} f(\lambda) = 0$.

So we have

$$\lim_{\lambda \rightarrow 1^-} (1 - \lambda) v_\lambda = g$$

Markov Reward Process - Average Reward

Theorem - Relation with Total Reward

Define the expected total reward $v_N : \mathcal{S} \rightarrow \mathbb{R}$ after N steps as

$$v_N(s) = \mathbb{E} \left[\sum_{k=0}^N r(S_k) \mid S_1 = s \right].$$

Then

$$v_{N+1} = Ng + h + f(\lambda)$$

where $\lim_{\lambda \rightarrow 1^-} f(\lambda) = 0$.

Interpretation, Draw on the board

Markov Reward Process - Average Reward

Average Reward per Stage Problem

$$V(s) = \mathbb{E} \left[\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T R(S_t) \mid S_1 = s \right]$$

Difficulties:

- Not necessary contractive
- All time steps as important (slow convergence)
- Others

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Value iteration for an average-reward MRP

Consider a Markov reward process $(\mathcal{S}, P, r)$ and define

$$L(h) = r + Ph.$$

Starting from any $h_0 \in \mathbb{R}^{|\mathcal{S}|}$, value iteration gives

$$h_{k+1} = L(h_k) = r + Ph_k.$$

The gain g^* and a bias h^* satisfy

$$g^* \mathbf{1} + h^* = r + Ph^*.$$

Important

Unlike discounted value iteration, h_k does not generally converge.

Value Iteration

To study the convergence of this VI, we consider the span semi-norm:

$$\text{sp}(v) = \max_{s \in \mathcal{S}} v(s) - \min_{s \in \mathcal{S}} v(s)$$

The seminorm has some properties:

- $\text{sp}(v) \geq 0$ for all v
- $\text{sp}(v + w) \leq \text{sp}(v) + \text{sp}(w)$ (Triangle inequality)
- $\text{sp}(cv) = |c|\text{sp}(v)$ for all scalars c
- $\text{sp}(v + ce) = \text{sp}(v)$ for all scalars c
- $\text{sp}(v) = \text{sp}(-v)$
- $\text{sp}(v) \leq 2\|v\|$

Value Iteration

- Norms measure how far a vector is from zero.
- The span semi-norm measure, $\text{sp}(v) = 0$ does not imply that $v = 0$.
- The span semi-norm measure how far a vector is from being all constant. ($\text{sp}(ce) = |c|\text{sp}(0) = 0$)

Value Iteration

Proposition

For transition matrix P , we have

$$\text{sp}(Pv) \leq \gamma \text{sp}(v)$$

where

$$\gamma = 1 - \min_{s,z \in \mathcal{S}} \min_{s' \in \mathcal{S}} \{P(s'|s) - P(s'|z)\}$$

$\gamma = 1$ if at least two rows of P are orthogonal, otherwise less than 1.

What converges?

Assume that, for some $\gamma < 1$,

$$\text{sp}(Pv) \leq \gamma \text{sp}(v) \quad \text{for all } v.$$

Then

$$\text{sp}(h_k - h^*) \leq \gamma^k \text{sp}(h_0 - h^*).$$

Thus, up to an additive constant, $h_k \rightarrow h^*$. However, the constant component drifts approximately as

$$h_k \approx kg^*\mathbf{1} + h^* + c\mathbf{1}.$$

Consequently,

$$h_k \begin{cases} \rightarrow +\infty, & g^* > 0, \\ \rightarrow -\infty, & g^* < 0, \end{cases} \quad \text{but} \quad h_{k+1} - h_k \rightarrow g^*\mathbf{1}.$$

Homework 2

Prove geometric of $h_{k+1} - h_k$ to the gain.

Relative value iteration

The divergence occurs only in the direction of $\mathbf{1}$. Choose a reference state $\bar{s}$ and remove the constant component:

$$\hat{h}_{k+1} = L(\hat{h}_k) - L(\hat{h}_k)(\bar{s})\mathbf{1}.$$

Therefore,

$$\hat{h}_{k+1}(\bar{s}) = 0 \quad \text{at every iteration.}$$

Under the span-contraction assumption,

$$\hat{h}_k \longrightarrow h^* - h^*(\bar{s})\mathbf{1}.$$

Moreover, the removed constant gives an estimate of the gain:

$$g_k = L(\hat{h}_k)(\bar{s}) \longrightarrow g^*.$$

Main idea

Ordinary VI accumulates the gain at every iteration. Relative VI

Relative value iteration

Similar convergence results can be established generally for both value iteration and the relative value iteration, given that the underlying Markov chain is recurrent.

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Markov Decision Process

A Markov Decision Process (MDP) is a tuple $(\mathcal{S}, \mathcal{A}, P, r)$ of

- 1 **States:** $\mathcal{S}$
- 2 **Actions:** $\mathcal{A}$
- 3 **Transitions:** $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ where

$$\sum_{s' \in \mathcal{S}} P(s'|s, a) = 1 \quad \text{for all } (s, a) \in \mathcal{S} \times \mathcal{A}.$$

- 4 **Reward function:** $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$

MDP - Trajectory

A trajectory (realization) of $(\mathcal{S}, P, R)$ is a sequence

$$S_0, A_0, R_0, S_1, A_1, R_1, S_2, A_2, R_2 \dots, S_t, A_t, R_t \dots$$

where

$$\begin{aligned} P(S_{t+1}=s' | S_t=s, A_t=a_t, \dots, A_0=a_0) &= P(S_{t+1}=s' | S_t=s, A_t=a) \\ &= P(s, s') \end{aligned}$$

and

$$R_t = R(S_t, A_t) \quad [\text{or } R_t \sim R(S_t, A_t)]$$

MDP - Policy

Want to find a good policy:

$$\pi : \mathcal{S} \rightarrow \mathcal{A} \quad (\text{Deterministic})$$

$$p_{\pi}(a|s) : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1] \quad (\text{Stochastic})$$

For a fixed policy π we have a MRP with transition matrix P_{π} where

$$P_{\pi}(s, s') = P(s'|s, \pi(s))$$

MDP - Value

$$V_{\pi}(s) = \left[\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T R(S_t) \mid S_1 = s \right]$$

Can we find the optimal policy?

$$\max_{\pi} V_{\pi}(s)$$

MDP - Value of a State

Can we find the optimal policy?

$$\max_{\pi} V_{\pi}(s)$$

Assumption

For every policy π , the Markov chain $(\mathcal{S}, P_{\pi})$ has only one recurrent class.

Optimality Conditions

Theorem - Optimality condition

There exists a unique gain g^* and a basis function $h^* : \mathcal{S} \rightarrow \mathbb{R}$ such that for any $s \in \mathcal{S}$

$$g^* + h^*(s) = \max_{a \in \mathcal{A}} r(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a) h^*(s').$$

The policy

$$\pi(s) = \operatorname{argmax}_{a \in \mathcal{A}} r(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a) h^*(s')$$

is optimal.

Value Iteration

Idea iteratively apply the optimal Bellman operator:

$$B(h)(s) = \max_{a \in \mathcal{A}} r(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a)h(s').$$

Value Iteration

Idea iteratively apply the optimal Bellman operator:

$$B(h)(s) = \max_{a \in \mathcal{A}} r(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a)h(s').$$

Value Iteration Algorithm

Initialize $h_0 \in \mathbb{R}^{|\mathcal{S}|}$. At iteration k , let h_k be given. Then for all $s \in \mathcal{S}$ perform the update

$$h_{k+1}(s) = \max_{a \in \mathcal{A}} r(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a)h_k(s').$$

When stopping criterion is met, return

$$\pi(s) = \operatorname{argmax}_{a \in \mathcal{A}} r(s, a) + \sum_{s' \in \mathcal{S}} P(s'|s, a)h(s')$$

Value Iteration

Similar convergence properties as for the MRP, and can also do relative value iteration.

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Average-reward optimality equation

The optimal gain and bias satisfy for every state s ,

$$g^* + h^*(s) = \max_a \left\{ r(s, a) + \sum_{s'} P(s'|s, a) h^*(s') \right\}.$$

Idea

Replace the maximum by one inequality for every action.

Primal LP: gain and bias

The Bellman inequalities are

$$g + h(s) \geq r(s, a) + \sum_{s'} P(s'|s, a)h(s'), \quad \forall(s, a).$$

Therefore, solve

$$\begin{aligned} & \underset{g, h}{\text{minimize}} && g \\ & \text{subject to} && g + h(s) \geq r(s, a) + \sum_{s'} P(s'|s, a)h(s'), \\ & && \forall(s, a). \end{aligned}$$

Since h is defined only up to a constant, we may impose

$$h(\bar{s}) = 0.$$

Dual LP: steady-state flows

Associate a nonnegative variable $x(s, a)$ with each state-action pair:

$$\begin{aligned}
 & \underset{x}{\text{maximize}} && \sum_{s,a} r(s, a)x(s, a) \\
 & \text{subject to} && \sum_a x(s, a) = \sum_{s',a} P(s \mid s', a)x(s', a), \quad \forall s, \\
 & && \sum_{s,a} x(s, a) = 1, \\
 & && x(s, a) \geq 0.
 \end{aligned}$$

Flow interpretation

For every state: long-run flow out = long-run flow in.

The normalization makes x a probability distribution over

Meaning of the dual variables

The variable $x(s, a)$ represents the long-run fraction of time that the process is in state s and chooses action a . Define the state occupancy

$$\mu(s) = \sum_a x(s, a).$$

A stationary policy can be recovered by

$$\pi(a | s) = \frac{x(s, a)}{\mu(s)} \quad \text{when } \mu(s) > 0.$$

Its average reward is

$$\sum_{s,a} r(s, a)x(s, a).$$

Two equivalent viewpoints

Gain and bias	$\iff$	Steady-state occupancy
(g^*, h^*)	$\iff$	x^*

At optimality, $x^*(s, a) > 0$ only for actions that are tight in the Bellman